Subproject 1: Faculty Interaction (by Raja Darshini Rajamani)

3.1. Research Question and Hypothesis

Faculty interaction measures how frequently and meaningfully students connect with their professors outside the usual class environment, including through mentorship, advising, or general support. It's important because strong relationships with faculty can boost students' confidence academically, guide their professional goals, and help them build practical skills. When students have positive experiences with professors, they're more likely to see coursework as relevant to their lives and careers, improving their motivation and engagement. By focusing on how factors like emotional support, mentorship, and career guidance affect these interactions, we can better understand what helps students thrive and use that knowledge to improve educational practices.

Variable	Description
Faculty Interaction (FAC_INTERACTION), Y1	Student – Faculty Interaction Score
Satisfaction with Courses (SATIS10_9406), X1	Satisfaction with courses in major field
Academic Advising (SATIS02_9406), X2	Satisfaction with Academic Advising
Tutorial Help (SATIS36_9406), X3	Satisfaction with Tutorial help or other academic assistance
Mentorship (SATIS01_9406), X4	Ability to find faculty or staff mentor
Relevance of coursework (SATIS31_9406), X5	Relevance of coursework to everyday life
Support & Encouragement (FACPRV08), X6	Emotional support and encouragement provided by faculty
Study Skills (FACPRV07), X7	Faculty helps in improving study skills
Career Help (FACPRV13), X8	Faculty helps in achieving professional goals

Full Model:

$$\beta_0 + \beta_1(Courses\ Satisfaction) + \beta_2(Academic\ Advising) + \beta_3(Tutorial\ Help) + \beta_4(Mentorship) + \beta_5(Relevance\ of\ Coursework) + \beta_6(Encouragement) + \beta_7(Study\ Skills) ++ \beta_8(Career\ Help) + \varepsilon$$

Research Question 1: To what extent do students' ability to find a mentor, encouragement from faculty, faculty help in improving study skills, and faculty assistance in achieving professional goals predict the level of faculty interaction?

H_0: $\beta_4 = \beta_6 = \beta_7 = \beta_8 = 0$ (None of these predictors significantly impact faculty interaction)

H_1: $At\ Least\ one\ of\ \beta_4, \beta_6, \beta_7, \beta_8 \neq 0$ (At least one predictor significantly impacts faculty interaction)

Research Question 2: Does satisfaction with classes interact with the relationship between faculty assistance in professional goals and faculty interaction?

H_0: $\beta_{1\times8} = 0$ (There is no interaction effect between satisfaction with classes and faculty assistance in achieving professional goals on faculty interaction)

H_1: $\beta_{1\times8} \neq 0$ (There is an interaction effect between satisfaction with classes and faculty assistance in achieving professional goals)

3.2. Methodology

We used a confirmatory workflow. First, an 8-predictor OLS established a baseline; VIFs showed no collinearity, but Cook's D and BP revealed leverage clusters and heteroskedasticity. WLS and robust M-estimation improved precision only marginally, so we trimmed the top 15 % high-influence rows, refitted, and applied Best-subset and stepwise AIC then identified a 7-predictor core, and a general linear test confirmed tutorial-help satisfaction was dispensable. A separate model validated the Courses Satisfaction × Career-Help interaction. Finally, 10-fold cross-validation (RMSE = 2.4, $R^2 = 0.91$) and a 1 500-draw bootstrap demonstrated stable out-of-sample performance.

3.3. Model Diagnostics and Remedies

The full model (8 variables) has an Adj-R2 of 83.39% and Residual SE of 3.474 with a df of 188868. All the variables show significance. The plot in (Figure A1) shows Non-Constant Variance (clear heteroskedasticity), Non-Normal Errors and influential points. A BP-Test confirms the heteroscedasticity (BP = 444.1, $p < 2 \times$

10^{-16}). The Type II Anova (FIGURE A2) test shows *Career Help* explains 2.7 M units of SS and has F = 225k. *Encouragement* and *Study Skills Help* are the next two heavy hitters. *Tutorial Help* contributes very little unique variance (SS = 88, F = 7.2, p = 0.007).

Multicollinearity: VIFs are well below the common concern thresholds (10). Tolerance values are all > 0.10, further confirming no warning signs.

Outlier and Influential Points: Cook's D points at a handful of large, labelled bubbles and high vertical spikes as high influence. Studentized Residual shows few points exceeding the Bonferroni $p < 0.05$ cut-off. The Hat Matrix points to a few outliers. DFBETAS shows that no individual observations swing any single β by more than about 0.03 units; global influence is modest (Figure A3).

WLS: The Residual SE improved from 3.474 to 1.255. For the coefficients, the signs remain unchanged; magnitudes move < 3 %. The BP test on WLS model shows **(**BP = 6 620, $p < 2 \times 10^{-16}$). Variance is still non-constant. BP grew in WLS compared to OLS because the weighting scheme over-corrected in places and under-corrected in others. The WLS plot didn't change compared to OLS.(Figure A4)

Robust: Huber and Bisquare, both coefficients were stable, all shifts were under 0.05 units. Residual SE stays at 3.48, robust didn't tighten the spread. Significances were unchanged. Outliers have only a marginal influence on slope estimates; OLS estimates are already robust enough for this data set. (Figure A5)

Trimming Data: Robust and WLS tweaks barely changed anything so rows exceeding any of the flags (|rstandard| > 2, |rstudent| > 3, leverage > 2 p/n, Cook's D > 50%, |DFBETAS| > $2/\sqrt{n}$) were deleted, eliminating the leverage cluster seen earlier. Residual standard error falls from 3.47 to 2.36; Adj-R^2 rises by eight points to 0.9155, meaning the model now explains 92 % of the cleaned-sample variance. Practical insignificance of ***Tutorial Help*** confirmed by the insignificance shown by p value. The new plot shows Heteroskedasticity remains. Residual band is roughly centred on zero, and the spread is far flatter than before. The QQ plot's centre aligns well, but with heavy tail (expected with large n). (Figure A6)

CR Plots: All eight residual vs X and CR plots lie virtually flat, with only negligible wiggles; none display systematic bends or fan-shapes. Hence the trimmed model's relationship with each predictor is effectively linear, and no quadratic or spline terms are needed. ((Figure A7)

Added Variable Plots: The *Tutorial Help* panel is a nearly flat strip with a slope = 0, meaning it contributes no additional explanatory power once the other predictors are in the model. Other variables slope upward, and the blue fit line is clearly non-zero, so each of these variables still adds linear information to explain Faculty Interaction. (Figure A8)

Transformation on Y (BoxCox): The Box-Cox transform (λ = 0.55) almost flattens the residual-vs-fitted stripe and cuts the BP statistic from 4 812 to 1 179, but with n=137k even that reduced heteroskedasticity is still flagged. Adding WLS on top of the transformed response shaves BP further to 311 and drives the residual SE down to 0.031 while leaving all coefficient signs and their relative magnitudes unchanged.(Figure A9)

Bootstrap: (1 500-draw) Trimming plus bisquare weighting gives stable coefficients; bootstrap CIs reiterate that every predictor except *Tutorial Help*, CI (-0.018, 0.003) has a firmly positive and practically meaningful effect on faculty interaction. (Figure A10)

Best Subset: Best-subset search confirms that a 7-predictor model is the sweet spot. Moving from 6 to 7 variables lifts Adj-R^2 to 0.9155, but adding the 8th variable makes Cp, BIC, AICc and PRESS all worse. The optimal set remains *Courses Satisfaction, Academic Advising, Mentorship, Relevance of Coursework, Encouragement, Study Skills, Career Help*, exactly the full trim minus the now‑dispensable *Tutorial Help*.

Stepwise: Stepwise AIC confirms the earlier best-subset result: dropping *Tutorial Help* gives the best model while cutting the residual SE to 2.36. Alln7 remaining coefficients are still highly significant. The non variance still exists. (Figure A11)

KFold - 10-fold cross-validation again singles *Mentorship, Encouragement, Study Skills Help, Career Help* as the most reliable out-of-sample predictors. Across ten held-out folds the model averages RMSE = 2.40**,** MAE = 1.97, and R^2 = 0.912**,** essentially matching the in-sample trimmed fit, so its performance generalizes. Each coefficient remains positive, substantial, and interpretable, confirming that *mentoring, emotional support, study-skills coaching, and career guidance are the four faculty practices that matter most for boosting student–faculty interaction.*

3.4. Results and Hypothesis Testing

Research Question 1 – Main Effects of Four Faculty-Support Variables

The trimmed reduced model with Mentorship, Emotional support, Study-skills help, and Career help explains 91 % of the variance (Adj-$R^2 = 0.912$) with a residual SE of 2.40. Each coefficient is large and sharply estimated ($|t| \geq 94$; $p < 2 \times 10^{-6}$). Diagnostic plots still show mild heteroskedasticity (slight widening at the ends) and heavy tails, but no serious curvature or leverage problems remain. The F-test gives $F(4, 137\,236) = 3.56 \times 10^5$, $p < 0.001$, confirming all four predictors contribute unique information. **Conclusion → Reject H_0.** Mentorship, encouragement, study-skills coaching and career guidance each exert a strong, independent, and practically important impact on faculty-interaction scores. This aligns with earlier studies showing relationship-building resources matter more than purely instructional ones. (Figure A12)

Research Question 2 – Interaction of Course Satisfaction × Career Help

Interaction term $\beta_{1\times8} = 0.274$ ($t = 13.6$, $p < 2 \times 10^{-16}$). Partial-F versus the additive model: $F(1, 137\,237) = 184$, $p < 0.001$. **Conclusion → Reject H_0.** The positive slope means the career-help benefit is largest for students who are already happy with their classes, consistent with the motivational theory that satisfied students are more receptive to mentoring about future goals. Diagnostics look similar to earlier models, mild heteroskedasticity and heavy tails remain, but no new linearity or leverage issues appear. (Figure A13)

Takeaway: The interaction model by itself fits poorly (Adj-R^2 0.78, huge Cp/PRESS); the interaction is meaningful only when the strong main-effect variables are also present. The 4-predictor model is the leanest ($Cp = p = 5$) and its PRESS error is only 4 % higher than the 7-predictor. The 7-predictor model shaves an extra 0.35 percentage-points of Adj-R^2 and halves PRESS, at the cost of three more classroom-satisfaction variables. If focus on faculty-provided resources matter, choose the 4-predictor model; if cross-validated error is the priority, keep the 7-predictor main-effects model (interaction doesn't improve PRESS once the main effects are in).

Major Contribution - This study quantifies how strongly four specific faculty behaviors such as mentoring, emotional encouragement, study-skills coaching, and career guidance drive student–faculty interaction and demonstrates that course satisfaction further amplifies the career-help effect. These findings isolate clear, actionable levers for campuses looking to boost engagement. Some limitations and future directions are the Self-report bias may inflate both satisfaction and interaction scores; augmenting surveys with behavioural logs (e-mails, office-hour visits) would improve validity. The residual heteroskedasticity suggests modelling the variance structure (e.g., mixed-effects by department) for finer-grained insight.

Predictive Reach- Ten-fold cross-validation shows the lean four-predictor model averages $R^2 = 0.91$ and RMSE = 2.4, but that accuracy applies only to students resembling the 2023 survey cohort. Two factors curb out-of-sample power:

1. Sampling bias: respondents to lengthy climate surveys are typically more engaged, so predictions may overstate interaction levels for less-engaged or non-responding students and for other institutions.
2. Temporal drift: shifts in mentoring programs, advising formats, or campus culture can quickly render today's coefficients stale, inflating future error.

Hence, while k-fold CV confirms strong internal consistency, robust external prediction demands fresh, representative data and periodic model recalibration. Addressing these points will extend the present findings and help translate them into targeted, evidence-based faculty-development programmes.

Discussion

Conclusion

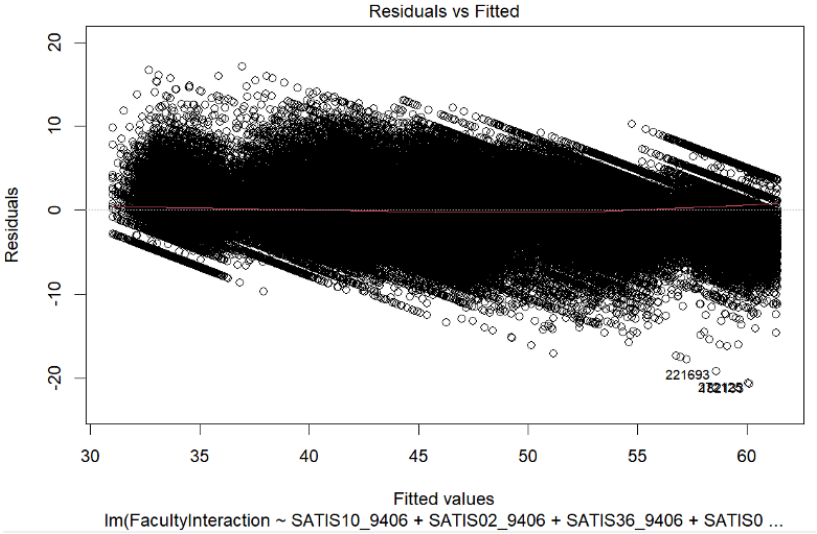
Our project collectively explored the multifaceted factors influencing undergraduate academic performance. Across different perspectives—self-perception, time allocation, faculty support, social development, and institutional services—we found that academic success is not determined by a single factor but by a complex interplay of personal, behavioral, and environmental factors. Students' confidence in their intellectual and writing abilities, the way they distribute time between study and extracurricular activities, the quality of faculty guidance and mentoring, their level of social engagement, and the support provided by campus services all meaningfully contribute to student academic outcomes. These findings highlight that both individual behaviors and institutional structures play critical roles in shaping students' educational journeys and broader development.

To foster stronger academic and personal outcomes, institutions should encourage students to build on their academic strengths while maintaining balanced time management between study and personal activities. Strengthening faculty-student relationships through mentoring and support programs, expanding social engagement opportunities, and ensuring equitable access to student services such as housing and mental health resources are essential. Additionally, future efforts should recognize that the relationships between these factors and academic outcomes may not be purely linear, suggesting the need for more flexible, tailored interventions that support students' diverse needs and promote both academic excellence and personal growth.

References

Appendix

Figure A1:



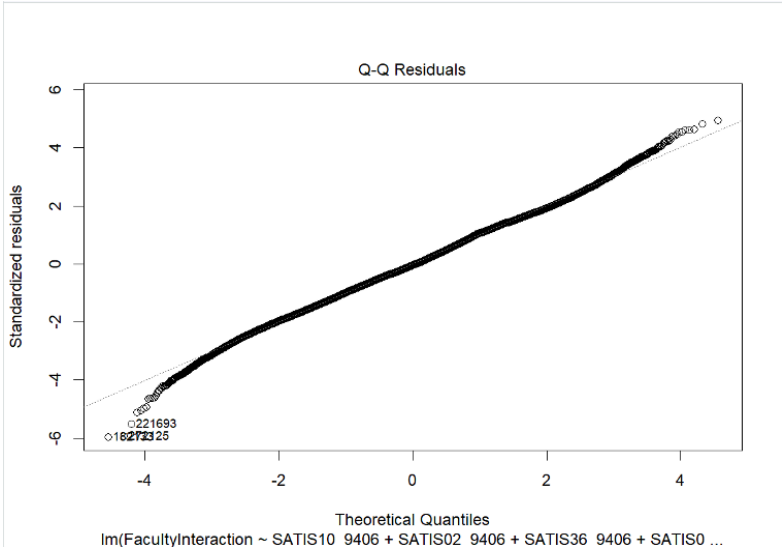
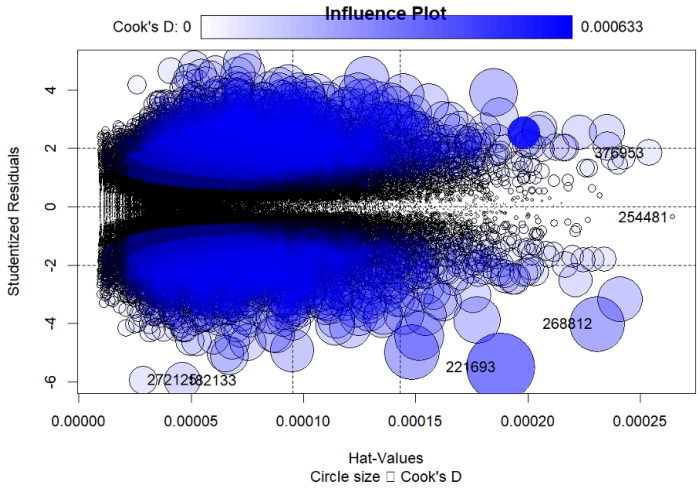


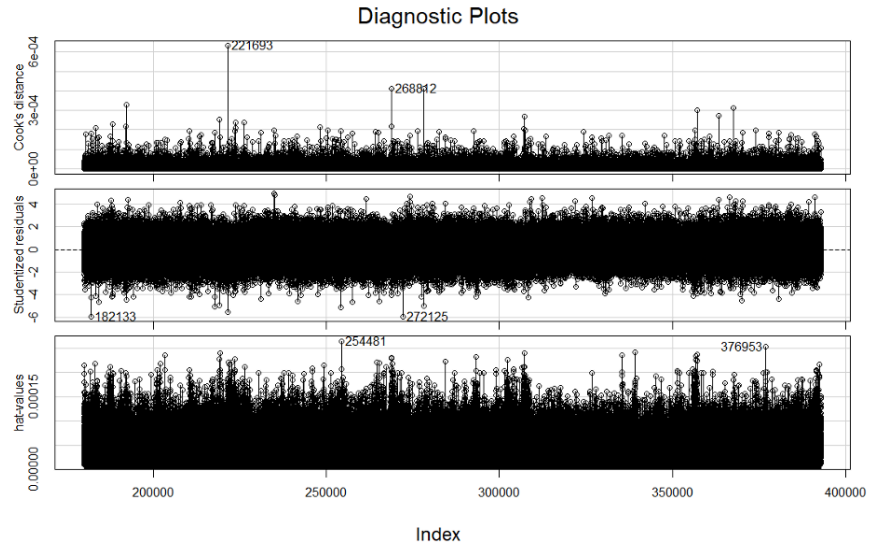
Figure A2:

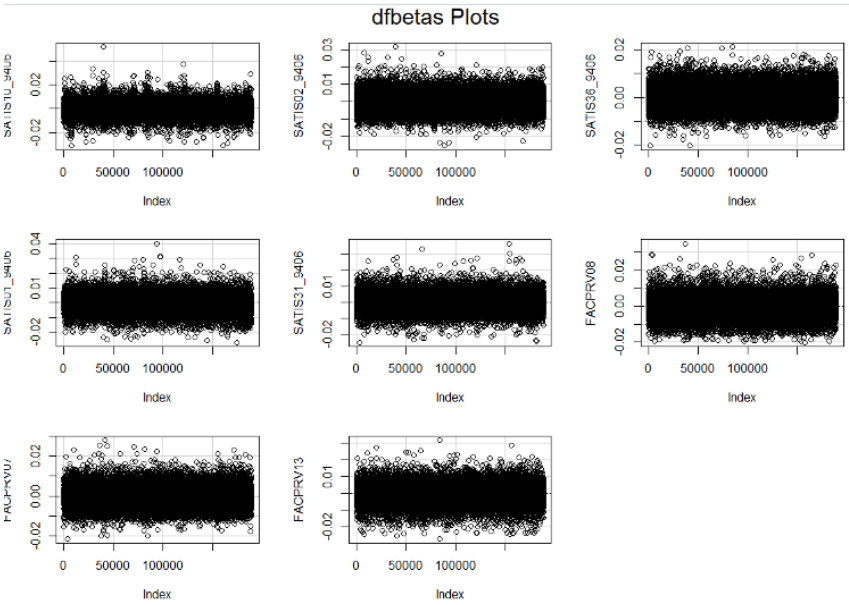
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> Anova(model, type = "II")
Anova Table (Type II tests)

Response: FacultyInteraction
              Sum Sq     Df    F value    Pr(>F)
SATIS10_9406   18052      1 1.4954e+03 < 2.2e-16 ***
SATIS02_9406   13588      1 1.1256e+03 < 2.2e-16 ***
SATIS36_9406      88      1 7.2492e+00  0.007094 **
SATIS01_9406   39681      1 3.2872e+03 < 2.2e-16 ***
SATIS31_9406    1892      1 1.5671e+02 < 2.2e-16 ***
FACPRV08     1168453      1 9.6793e+04 < 2.2e-16 ***
FACPRV07       87463      1 7.2453e+03 < 2.2e-16 ***
FACPRV13     2714669      1 2.2488e+05 < 2.2e-16 ***
Residuals    2279943 188868
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
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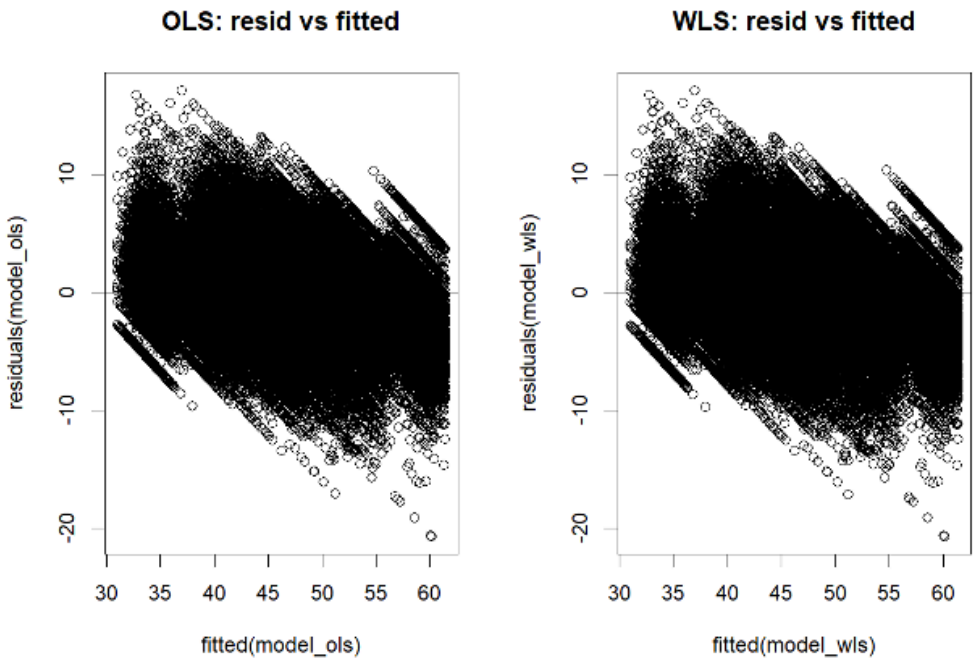
FigureA3:


Influence Plot
Cook's D: 0
0.000633
Studentized Residuals
Hat-Values
Circle size □ Cook's D
376953
254481
268812
221693
272125
182133


Diagnostic Plots
Cook's distance
Studentized residuals
hat-values
Index
221693
268812
182133
272125
254481
376953


dfbetas Plots
SATIS02_9406
SATIS36_9406
SATIS31_9406
FACPRV08
FACPRV13
Index

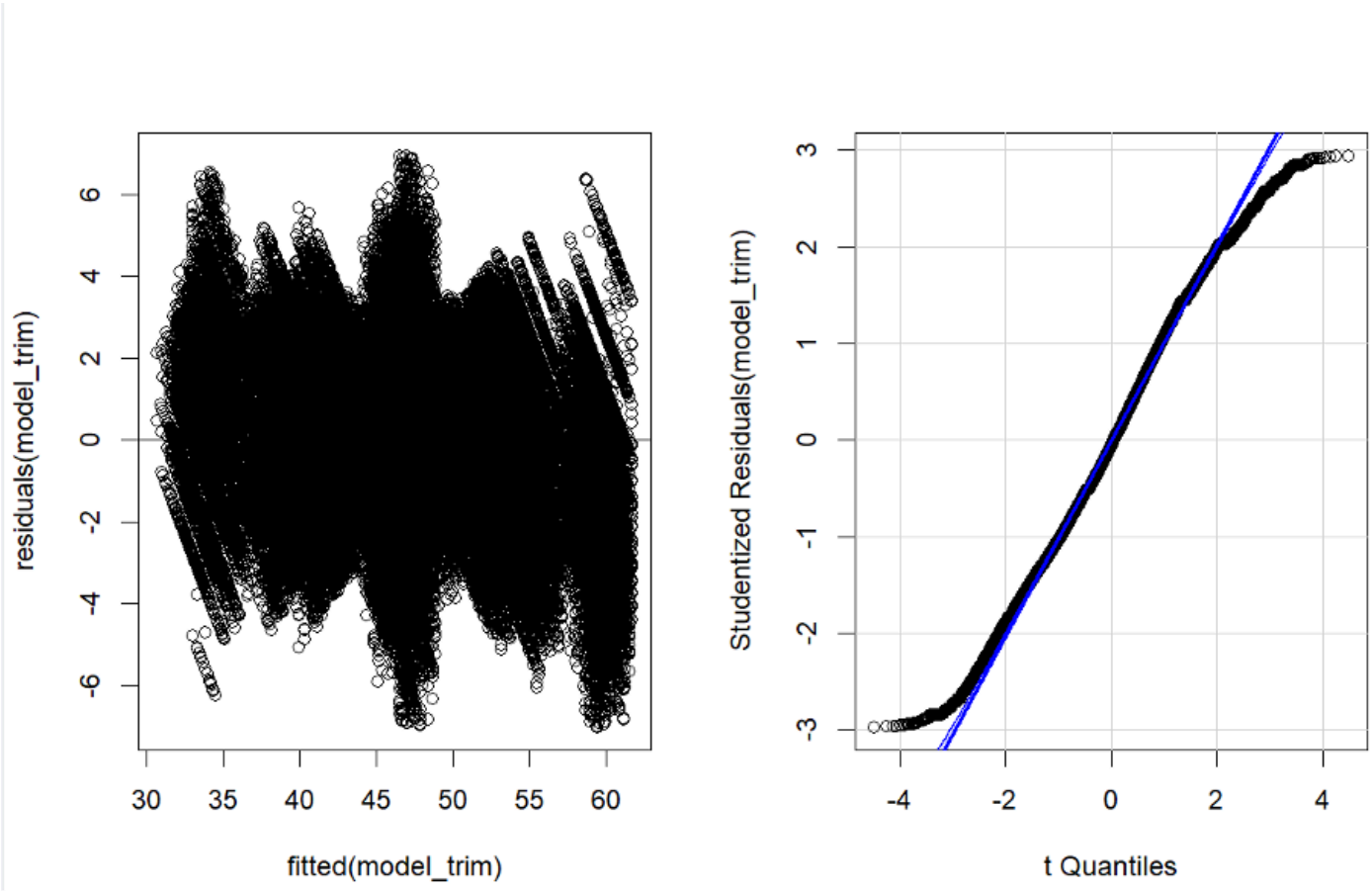
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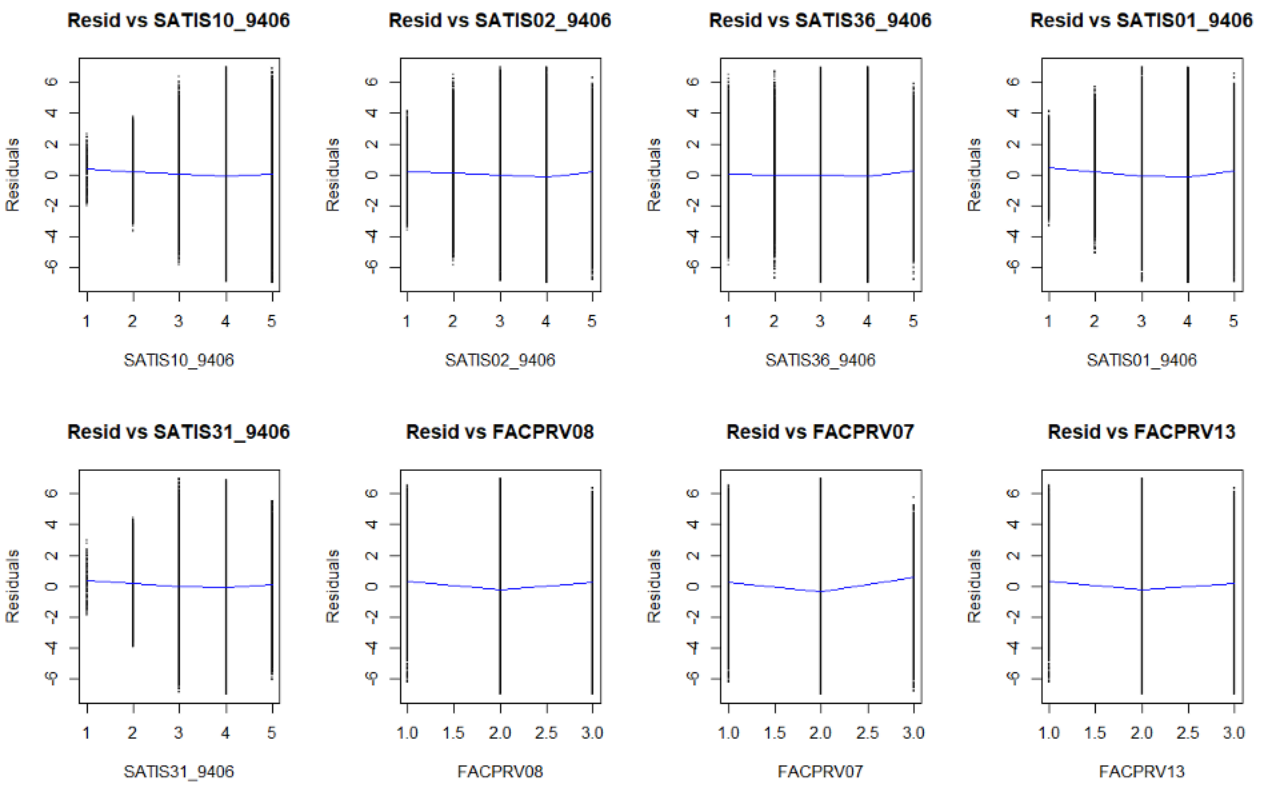
FigureA5:

Term	*OLS β̂*	*Huber β̂*	*Bisq β̂*	*Δ Huber–OLS (%)*	*Δ Bisq–OLS (%)*
(Intercept)	*17.1204*	*16.7427*	*16.7969*	*–2.2 %*	*–1.9 %*
SATIS10_9406	*0.3806*	*0.3930*	*0.3905*	*+3.2 %*	*+2.6 %*
SATIS02_9406	*0.2904*	*0.2964*	*0.2932*	*+2.1 %*	*+1.0 %*
SATIS36_9406	*–0.0175*	*–0.0179*	*–0.0198*	*–2.3 %*	*+12.8 %*
SATIS01_9406	*0.4855*	*0.4923*	*0.4885*	*+1.4 %*	*+0.6 %*
SATIS31_9406	*0.1334*	*0.1431*	*0.1399*	*+7.3 %*	*+4.8 %*
FACPRV08	*4.5026*	*4.5406*	*4.5389*	*+0.8 %*	*+0.8 %*
FACPRV07	*1.1556*	*1.2000*	*1.1822*	*+3.8 %*	*+2.3 %*
FACPRV13	*6.9569*	*6.9985*	*7.0103*	*+0.6 %*	*+0.8 %*

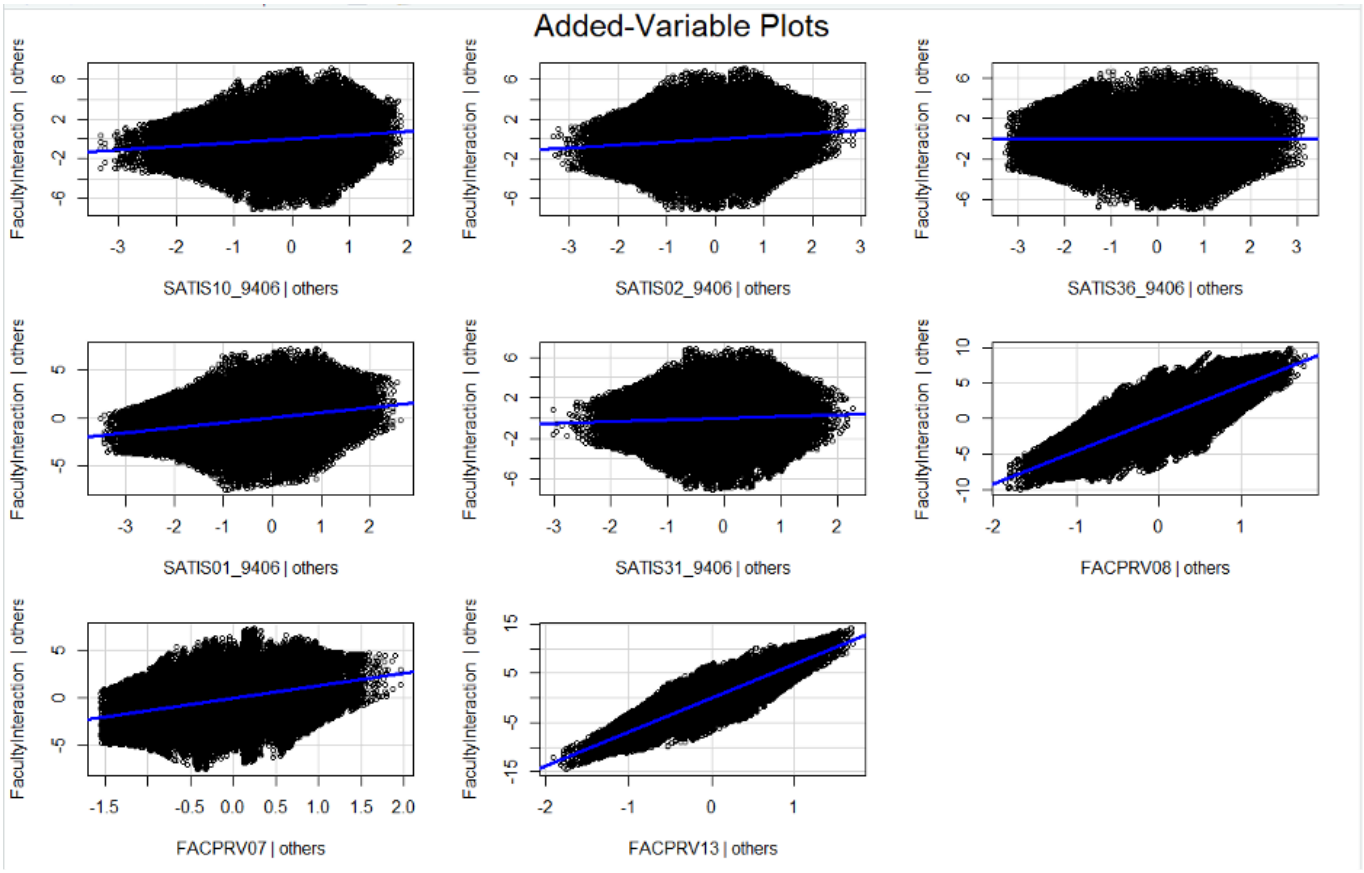
FigureA6:


residuals(model_trim)
fitted(model_trim)
Studentized Residuals(model_trim)
t Quantiles

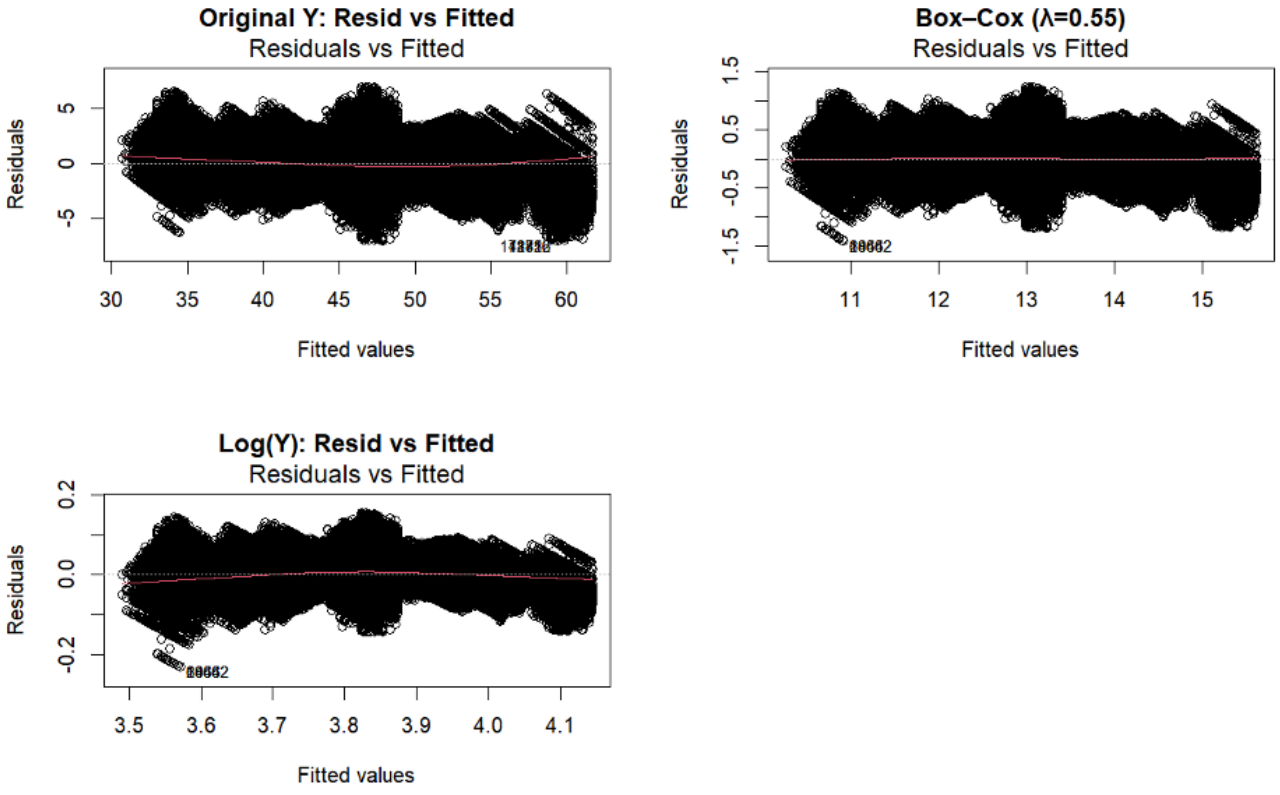
FigureA7:


Resid vs SATIS10_9406
Resid vs SATIS02_9406
Resid vs SATIS36_9406
Resid vs SATIS01_9406
Resid vs SATIS31_9406
Resid vs FACPRV08
Resid vs FACPRV07
Resid vs FACPRV13
Residuals
SATIS10_9406
SATIS02_9406
SATIS36_9406
SATIS01_9406
SATIS31_9406
FACPRV08
FACPRV07
FACPRV13

FigureA8:



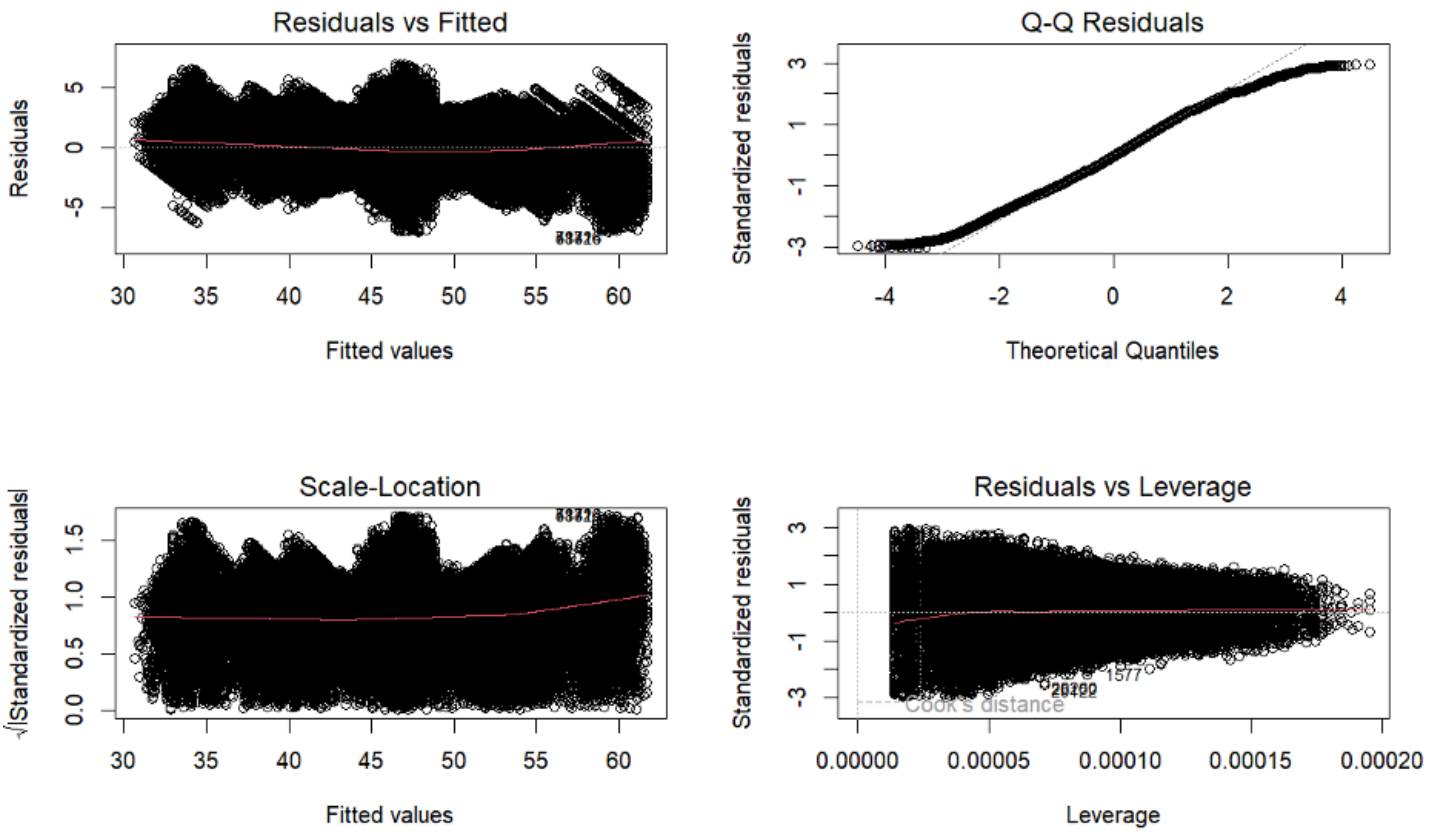
FigureA9:



FigureA10:

Term (predictor)	*2.5 %*	*97.5 %*
Intercept	*16.29*	*16.46*
SATIS10_9406 – Course Satisfaction	*0.371*	*0.405*
SATIS02_9406 – Academic Advising	*0.282*	*0.310*
SATIS36_9406 – Tutorial Help	*–0.018*	*0.003*
SATIS01_9406 – Mentor Ability	*0.502*	*0.531*
SATIS31_9406 – Coursework Relevance	*0.157*	*0.194*
FACPRV08 – Emotional Encouragement	*4.558*	*4.608*
FACPRV07 – Study-Skills Coaching	*1.254*	*1.301*
FACPRV13 – Career Guidance	*6.912*	*6.961*

FigureA11:



FigureA12:

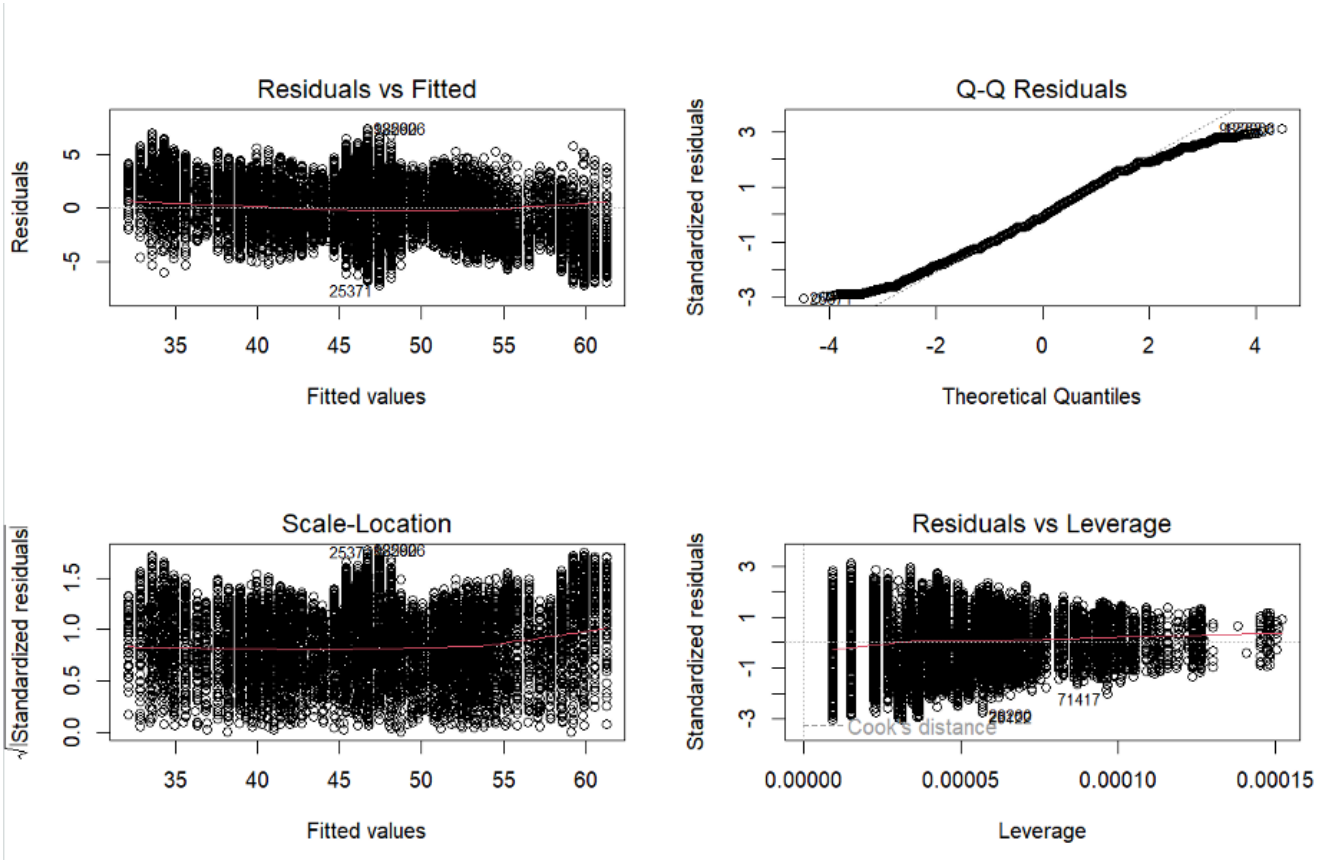
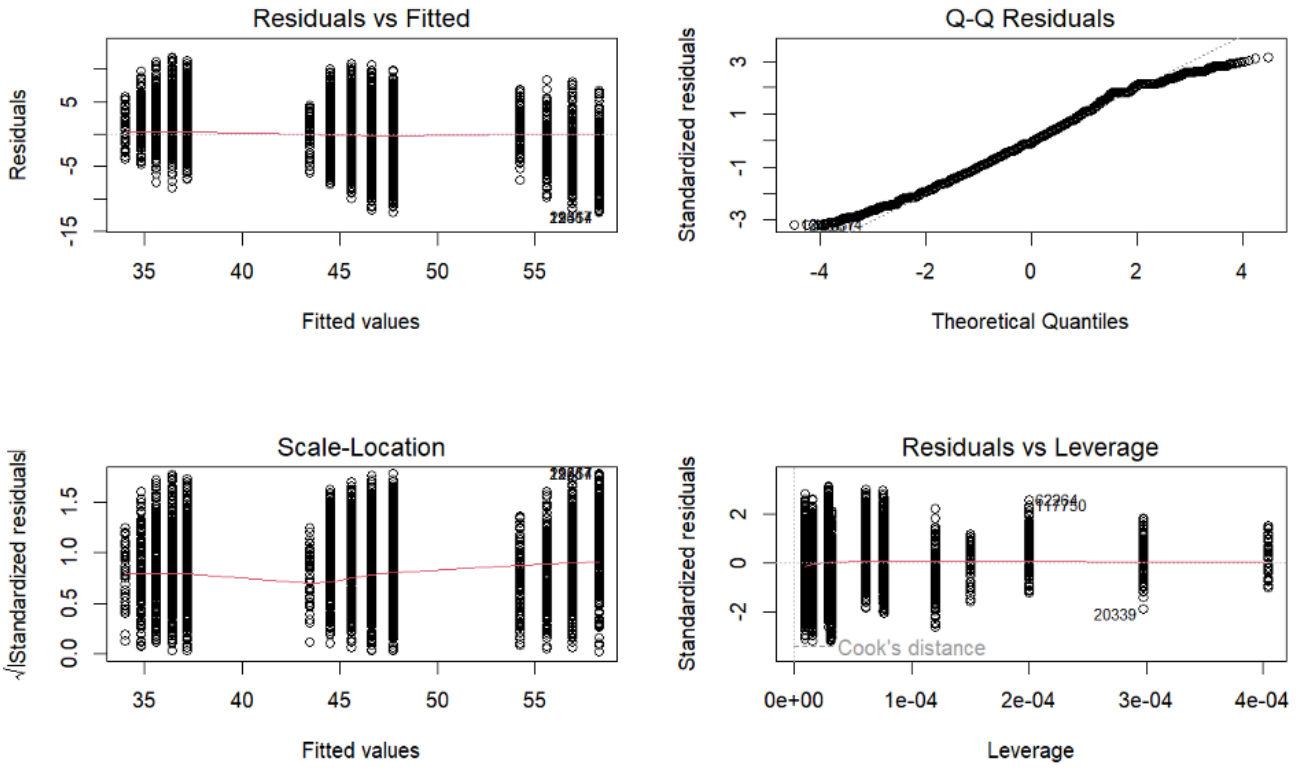

Residuals vs Fitted
Residuals
Fitted values
Q-Q Residuals
Standardized residuals
Theoretical Quantiles
Scale-Location
√|Standardized residuals|
Fitted values
Residuals vs Leverage
Standardized residuals
Cook's distance
Leverage

Figure A13:


Residuals vs Fitted
Residuals
Fitted values
Q-Q Residuals
Standardized residuals
Theoretical Quantiles
Scale-Location
√|Standardized residuals|
Fitted values
Residuals vs Leverage
Standardized residuals
Cook's distance
Leverage